

AOS-CX 10.07 CoPP Guide

6100 Switch Series



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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- Aruba 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
example-text	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ■ <code><example-text></code> ■ <code><example-text></code> ■ <i>example-text</i> ■ <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. ■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.

Convention	Usage
... or ...	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment. The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch (CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the `interface` context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where `<VLAN-ID>` is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

```
member/slot/port
```

On the 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 on the switch.

Overview

CoPP provides a way for administrators to protect the management processor on the switch from high packet loads (generated by malicious or nonmalicious sources) that might interfere with its ability to keep data plane traffic flowing. For example, a denial of service attack can result in excessive traffic that would slow down the management processor and negatively affect switch throughput.

A CoPP policy is composed of one or more classes. Each class defines one or more target protocols and how their traffic is managed. Every policy also has a default class to regulate packets that do not match any other class. The following actions can be applied for all packets matching a class:

- Drop the packets. (Excluding the default class.)
- Set the processing priority in the range 0 to 6 (0 - highest priority and 6 - lowest priority).
- Set the maximum data rate in packets per second (pps) at which each line module can send packets to the management processor.

Up to 32 CoPP policies can be defined, but only one can be active on the switch at a time.

A CoPP policy must always be active on the switch. By default, the switch has a CoPP policy named **default** which is automatically applied at first boot.

When the switch is rebooted, the CoPP policy that was actively applied to the switch before the reboot occurred will be applied if it was saved to the startup configuration with the `copy running-config startup-config` command.

Configuring CoPP

Procedure

1. Configure the default CoPP policy, edit an existing policy, or create a policy with the command `copp-policy`.
2. Add, edit, or remove classes in the policy with the command `class`.
3. If the policy is not the active policy on the switch, apply it with the command `apply copp-policy`. (Changes made to an active policy take effect immediately and do not need to be applied.)
4. Review the CoPP policy configuration settings with the command `show copp-policy`.

Example

This example creates the following configuration:

- Defines a new policy named **My_CoppPolicy**.
- Adds two classes to the policy.
- Activates the policy.
- Displays policy configuration settings.

```

switch(config)# copp-policy My_CoppPolicy
switch(config-copp)# class igmp priority 6 rate 5000
switch(config-copp)# class lacp priority 2 rate 2000
switch(config-copp)# exit
switch(config)# apply copp-policy My_CoppPolicy
switch(config)# exit
switch# show copp-policy My_CoppPolicy
class                drop priority rate pps hardware rate pps
-----
igmp                  6          5000   5000
lacp                  2          2000   2000
default               1          6000   6000

```

Actual rates in hardware

Currently, the actual rate in hardware is determined by a mapping based on the configured rate, shown by the following table of the first nine actual rates.

The preceding table shows the first nine actual rates available in hardware. Higher rates are available. The 'show copp-policy statistics class <CLASS>' command can be used to show the actual rate in hardware for a specific class. The 'show copp-policy <NAME>' command can be used to show the actual rates for all classes configured in a specific CoPP policy, if the policy is actively applied.

CoPP commands

Classes of traffic

The different classes of traffic that can be individually configured are:

- **arp-broadcast**: Address Resolution Protocol packets with a broadcast destination MAC address.
- **arp-unicast**: Address Resolution Protocol packets with a switch system destination MAC address.
- **dhcp**: Dynamic Host Configuration Protocol packets. Also includes snooped DHCP packets if DHCP snooping is enabled.
- **icmp-broadcast-ipv4**: Internet Control Message Protocol packets with a broadcast or multicast destination IPv4 address.
- **icmp-multicast-ipv6**: Internet Control Message Protocol packets with a well-known multicast destination IPv6 address.
- **icmp-security-ipv6**: IPv6 Internet Control Message Protocol packets intercepted and inspected.
- **icmp-unicast-ipv4**: Internet Control Message Protocol packets with a destination IPv4 address owned by the switch
- **icmp-unicast-ipv6**: Internet Control Message Protocol packets with a destination IPv6 address owned by the switch.
- **igmp**: Internet Group Management Protocol packets.
- **lacp**: Link Aggregation Control Protocol packets with the destination MAC address 01:80:c2:00:00:02.
- **lldp**: Link Layer Discovery Protocol packets with the destination MAC address 01:80:c2:00:00:0e.
- **loop-protect**: Loop Protection packets with the destination MAC address 09:00:09:09:13:a6.
- **manageability**: Unicast IP packets addressed to the switch for specific protocols that do not have a dedicated CoPP class like HTTP, SSH, RADIUS.
- **mvrp**: Multiple VLAN Registration Protocol packets with the destination MAC address 01:80:c2:00:00:20 or 01:80:c2:00:00:21

- `ntp`: Network Time Protocol packets with a destination IP address owned by the switch.
- `stp`: Spanning Tree Protocol (STP) packets with the destination MAC address 01:80:c2:00:00:00 or Per-VLAN Spanning Tree (PVST) packets with the destination MAC address 01:00:0c:cc:cc:cd.
- `udld`: Unidirectional Link Detection packets with the destination MAC address 01:00:0c:cc:cc:cc or 00:e0:52:00:00:00, or Cisco Discovery Protocol packets with the destination MAC address 01:00:0c:cc:cc:cc.
- `unresolved-ip-unicast`: Packets to be software forwarded by the management processor.

To regulate any other traffic destined for the CPU, every CoPP policy has a class named `default` that can also be configured to regulate other traffic to the CPU or prevent other traffic from being delivered.



All IPsec traffic received by the CPU will be regulated by the `ipsec` class regardless of the encapsulated protocol.

apply copp-policy

Syntax

```
apply copp-policy { <NAME> | default }
no apply copp-policy <NAME>
```

Description

Applies a CoPP policy to the switch, replacing the policy that is in effect. There may be a brief interruption in traffic flow to the management processor while the switch implements the change.

Enter the `no apply copp-policy <NAME>` command with the name of a CoPP policy to unapply a CoPP policy and apply the default CoPP policy. This will only take effect if the specified policy is actively applied. Since there must always be a CoPP policy applied, this command effectively attempts to replace the applied CoPP policy with the default CoPP policy. The default CoPP policy cannot be unapplied using this command.

Command context

`config`

Parameters

`<NAME>`

Specifies the name of the policy to apply. Length: 1 to 64 characters.

`default`

Applies the default policy.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If the new policy cannot be applied (for example, due to a lack of hardware resources), the previous policy remains in effect. Use the [show copp-policy](#) command to determine which policy is in effect.

Examples

Applying a policy named `My_CoppPolicy`:

```
switch(config)# apply copp-policy My_CoppPolicy
```

Applying the default policy:

```
switch(config)# apply copp-policy default
```

Unapplying a policy named My_CoppPolicy:

```
switch(config)# no apply copp-policy My_CoppPolicy
```

class

Syntax

```
class <CLASS> {drop | priority <PRIORITY> rate <RATE>}
```

```
no class <CLASS> {drop | priority <PRIORITY> rate <RATE>}
```

Description

Adds a class to a CoPP policy. If the class exists, the existing class is modified. Changes made to an active (applied) policy take effect immediately.

When adding or modifying a class in an active policy, CoPP immediately activates the change on the switch. In cases where insufficient hardware resources exist to support a class or its action, CoPP fails to activate the changed class on the switch. When this failure occurs, the active configuration on the switch will be out of sync with its definition. To diagnose and remedy this situation:

- Use the [show copp-policy](#) command to determine which classes are out of sync between the active policy and its definition.
- Use the [reset copp-policy](#) command to synchronize the active policy with its definition. This synchronization changes the classes in the definition to match the classes in the active policy.

The `no` form of this command removes the configuration for the class. Traffic for the class will be prioritized and regulated using the factory default configuration for the class. Use the `show copp-policy factory-default` command to display the factory default CoPP policy. To stop a class of traffic from reaching the processor, set the class action to drop.

Command context

config-copp

Parameters

<CLASS>

Specifies the class to add or edit.

drop

Drop packets matching the selected class.

priority <PRIORITY>

Specifies the priority for packets matching the selected class. Range: 0 to 6.

rate <RATE>

Specifies the maximum rate, in packets per second (pps), for packets matching the selected class. Range: 25 to 99999.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding a class to handle LACP traffic with priority of 2 and rate of 2000:

```
switch(config-copp) # class lacp priority 2 rate 2000
```

Modifying the class to drop LLDP packets:

```
switch(config-copp) # class lldp drop
```

Removing the class that handles LLDP packets.

```
switch(config-copp) # no class lldp
```

clear copp-policy statistics

Syntax

```
clear copp-policy statistics
```

Description

Resets statistics for all CoPP classes to zero.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying and then resetting statistics for all classes in the active policy:

copp-policy

Syntax

```
copp-policy {<NAME> | default [revert]}  
no copp-policy <NAME>
```

Description

Creates a CoPP policy and switches to the `config-copp` context for the policy. Or, if the specified policy exists, switches to the `config-copp` context for the policy. A predefined policy, named `default`, contains factory default classes and is applied to the switch at first startup. This policy cannot be deleted, but its configuration can be changed.

The `no` form of this command removes a CoPP policy. If a policy is active (applied), it cannot be removed. It must be replaced with another policy before it can be removed.

Command context

```
config
```

Parameters

<NAME>

Specifies the name of the policy to add or edit. Length: 1 to 64 characters. The name must not be a substring of any of the following reserved words: default, factory-default, commands, configuration, or statistics.

default

Specifies the default CoPP policy. Use this default policy to configure the default policy.

revert

Sets the default CoPP policy to its factory settings.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a policy named My_CoppPolicy:

```
switch(config)# copp-policy My_CoppPolicy
switch(config-copp)#
```

Removing a policy named My_CoppPolicy:

```
switch(config)# no copp-policy My_CoppPolicy
```

Setting the default policy to its factory settings:

```
switch(config)# copp-policy default revert
```

Unapplying the policy named My_CoppPolicy:

```
switch(config)# no apply copp-policy My_CoppPolicy
```

default-class

Syntax

```
default-class priority <PRIORITY> rate <RATE>
```

Description

Configures the default class that is automatically defined for all CoPP policies. The default class cannot be removed, but its configuration can be changed. The default class is applied to traffic that does not match any other class defined for a policy.

Command context

config-copp

Parameters

priority <PRIORITY>

Specifies the priority for packets matching the selected class. Range: 0 to 7.

rate <RATE>

Specifies the maximum rate, in packets per second (pps), for packets matching the selected class. Range: 25 to 99999.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the default class to a priority of **2** and rate of **2000**:

```
switch(config-copp) # default-class priority 2 rate 2000
```

reset copp-policy

Syntax

```
reset copp-policy { <NAME> | default }
```

Description

Resets an active CoPP policy to match the settings that are currently in effect for the active policy on the switch. Changes made to the active policy that could not be activated are removed from the active policy. When the switch fails to add or modify a class in an active CoPP policy, it is possible the active policy settings on the switch may be out of sync with those defined in the policy.

Command context

config

Parameters

<NAME>

Specifies the name of the policy to reset. Length: 1 to 64 characters.

default

Resets the default policy to match its active settings.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resetting a policy named My_CoppPolicy:

```
switch# show copp-policy My_CoppPolicy
class                drop priority rate pps hardware rate pps
-----
igmp                  6          5000   5000
lcp                   2          2000   2000
default               1          6000   6000
switch# config terminal
switch(config)# copp-policy My_CoppPolicy
switch(config-copp) # class stp priority 4 rate 4000
switch(config-copp) # do show copp-policy My_CoppPolicy
class                drop priority rate pps hardware rate pps
-----
```

```

igmp          6          5000      5000
lACP          2          2000      2000
default       1          6000      6000
% Warning: user-specified classes in CoPP policy My_CoppPolicy do not match
active configuration.
switch(config-copp)# do show copp-policy My_CoppPolicy configuration
class          drop priority rate pps applied
-----
igmp          6          5000      yes
lACP          2          2000      yes
stp           4          4000      no
default       1          6000      yes
% Warning: user-specified classes in CoPP policy My_CoppPolicy do not match
active configuration.
switch(config-copp)# exit
switch(config)# reset copp-policy My_CoppPolicy
switch(config)# do show copp-policy My_CoppPolicy
class          drop priority rate pps hardware rate pps
-----
igmp          6          5000      5000
lACP          2          2000      2000
default       1          6000      6000

```

Resetting the default policy:

```
switch(config)# reset copp-policy default
```

show copp-policy

Syntax

```
show copp-policy [<NAME> | default] [commands] [configuration]
```

Description

Shows CoPP policy settings for a specific CoPP policy. When entered without specifying either a name or the `default` parameter, shows all the CoPP policy settings that are active on the switch and have successfully been programmed into the hardware.

A warning is displayed if:

- The active and user-specified applications of a policy do not match.
- The active and user-specified configurations of a policy do not match.

Command context

Operator (>) or Manager (#)

Parameters

<NAME>

Specifies the name of the policy for which to display settings. Length: 1 to 64 characters.

`default`

Displays CoPP settings for the default policy.

`commands`

Displays output as CLI commands.

`configuration`

Displays user-specified CoPP settings and **not** the active settings.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Displaying the CoPP policies defined in the configuration and the active application:

```
switch# show copp-policy
applied copp_policy_name
-----
My_CoppPolicy
applied default
switch#
```

Displaying the active configuration of all CoPP policies as CLI commands:

Displaying the default policy:

```
switch# show copp-policy default
class                drop priority rate pps hardware rate pps
-----
arp-broadcast                2          1250      1250
arp-unicast                  3           825       825
  <--OUTPUT OMITTED FOR BREVITY-->
default                      2          4225      4225
```

show copp-policy factory-default

Syntax

```
show copp-policy factory-default [commands]
```

Description

Display the configuration for the factory-default CoPP policy.

Command context

Operator (>) or Manager (#)

Parameters

commands

Displays output as CLI commands.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Displaying the factory-default policy:

```
switch# show copp-policy factory-default
class                drop priority rate pps
```

```

-----
arp-broadcast          2          1250
arp-unicast            3          825
  <---OUTPUT OMITTED FOR BREVITY-->
default                2          4225

```

Displaying the active configuration of `My_CoppPolicy` (`My_CoppPolicy` is applied):

```

switch# config terminal
switch(config)# apply copp-policy My_CoppPolicy
switch(config)# do show copp-policy My_CoppPolicy
class                drop priority rate pps hardware rate pps
-----
igmp                  6          5000    5000
larp                  2          2000    2000
default               1          6000    6000

```

Displaying the active configuration of `My_CoppPolicy` as CLI commands:

```

switch# show copp-policy My_CoppPolicy commands
copp-policy My_CoppPolicy
  class igmp priority 6 rate 5000
  class larp priority 2 rate 2000
  default-class priority 1 rate 6000
apply copp-policy My_CoppPolicy

```

Displaying the user-specified configuration of `My_CoppPolicy`:

```

switch# show copp-policy My_CoppPolicy configuration
class                drop priority rate pps applied
-----
igmp                  6          5000    yes
larp                  2          2000    yes
default               1          6000    yes

```

Displaying the user-specified configuration of `My_CoppPolicy` as CLI commands:

```

switch# show copp-policy My_CoppPolicy commands configuration
copp-policy My_CoppPolicy
  class igmp priority 6 rate 5000
  class larp priority 2 rate 2000
  default-class priority 1 rate 6000
apply copp-policy My_CoppPolicy

```

show copp-policy statistics

Syntax

```
show copp-policy statistics [class <CLASS> | default-class | non-zero]
```

Description

Displays statistics for all classes, a single class, or all classes with non-zero statistics in the active CoPP policy.

Command context

Operator (>) or Manager (#)

Parameters

<CLASS>

Specifies the [class](#) for which to display statistics.

default-class

Displays statistics for the default class.

non-zero

Displays statistics for all classes with non-zero statistics.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Usage

If a single class is specified, the priority and rate that has been programmed in hardware for that class will be shown.

Examples

Applying the default CoPP policy and displaying statistics for all classes in the actively applied policy:



The rate displayed is the actual rate in hardware.

```
switch# config terminal
switch(config)# apply copp-policy default
switch(config)# exit
switch# show copp-policy statistics
Statistics for CoPP policy 'default':
Totals:
  packets passed      : 1000           packets dropped   : 1500
Class: default
  packets passed      : 400           packets dropped   : 600
Class: arp-broadcast
  packets passed      : 500           packets dropped   : 800
  <--OUTPUT OMITTED FOR BREVITY-->
```

Displaying statistics for the default class in the active policy:

```
switch(config)# show copp-policy statistics default-class
Statistics for CoPP policy 'default':
Class: default
Description: Default
  priority             : 2
  rate (pps)           : 4225

  packets passed       : 400           packets dropped     : 600
```

Displaying statistics for the class arp-broadcast in the actively applied policy:

```
switch# show copp-policy statistics class arp-broadcast
Statistics for CoPP policy 'default':
Class: arp-broadcast
Description: Address Resolution Protocol broadcast
  priority      : 2
  rate (pps)    : 1250

  packets passed : 500                packets dropped : 800
```

show tech copp

Syntax

```
show tech copp
```

Description

Displays the output of all show commands supported by CoPP.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Capturing the command output into a local file:

```
switch# show tech copp local-file
Show Tech output stored in local-file. Please use 'copy show-tech local-file' to
copy-out this file.

switch# copy show-tech local-file ?
  REMOTE_URL    URL of syntax
                 {tftp://|sftp://USER@}{IP|HOST}[:PORT][;blocksize=VAL]/FILE
  STORAGE_URL  URL of syntax usb:/file
switch# copy show-tech local-file
```

Accessing Aruba Support

Aruba Support Services	https://www.arubanetworks.com/support-services/
Aruba Support Portal	https://asp.arubanetworks.com/
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
Software licensing	https://lms.arubanetworks.com/
End-of-Life information	https://www.arubanetworks.com/support-services/end-of-life/
Aruba software and documentation	https://asp.arubanetworks.com/downloads

Accessing Updates

You can access updates from the Aruba Support Portal or the HPE My Networking Website.

Aruba Support Portal

<https://asp.arubanetworks.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>

Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://asp.arubanetworks.com/notifications/subscriptions> (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

Documentation Feedback

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